Dynamic Matrix Dominance

Time Limit: 1.0 seconds **Memory Limit:** 256 megabytes

You are given a dynamically growing sequence of N weighted tasks, where the i^{th} task has a primary weight A_i ($1 \le A_i \le 10^9$) and a secondary priority i.

For any non-empty prefix of tasks $T^{(k)} = \{1, 2, \ldots, k\}$, we construct a directed Bipartite Interaction Graph $G_k = (V_L, V_R, E)$ as follows:

- **Left Vertices** (V_L): Represents the indices $\{1, 2, \ldots, k\}$.
- **Right Vertices** (V_R): Represents all distinct values present in the set $\{A_1, A_2, \ldots, A_k\}$.
- **Edges** (E): For each task $j \in V_L$, there is a directed edge $j \to A_j$ with weight A_j.

A **Dominating Configuration G'** of graph G_k is defined as a subset of edges $E' \subseteq E$ such that every right vertex $v \in V_R$ has an in-degree of at least 1 in E'. The total cost of a configuration G' is the sum of the weights of all selected edges in E'.

For each k from 1 to N, compute the minimum cost of a Dominating Configuration for G_k.

Input Format

- The first line contains an integer T ($1 \le T \le 10^4$) — the number of test cases.
- For each test case:
  - The first line contains an integer N ($1 \le N \le 2 \times 10^5$) — the number of tasks.
  - The second line contains N space-separated integers $A_1, A_2, \ldots, A_N$ ($1 \le A_i \le 10^9$) — the primary weights of the tasks.

It is guaranteed that the sum of N over all test cases does not exceed 2×10^5.

Output Format

For each test case, print a single line containing N space-separated integers, where the k^{th} integer represents the minimum cost of a Dominating Configuration for G_k.

Sample 1

Input:

```
2
5
1 2 2 3 1
4
5 5 5 5
```

Output:

```
1 3 3 6 6
5 5 5 5
```

Note

In the first test case:

- *k*=3: V_L = {1, 2, 3}, V_R = {1, 2}. $E = \{(1,1),(2,2),(3,2)\}$. For Configuration G’, we will take the edges $\{(1,1),(2,2)\}$ only. We will not take the edge (3,2) because the vertices V_R already have at least 1 in-degree. So, the cost will be 1+2=3. This is the minimum possible cost.